SMARTFACTORY – OUTPUT SCREENSHOTS

Screenshot 1: Factory Dashboard

```
CHOICE 01 – Factory Dashboard
Enter your choice: 1

==========================================
           FACTORY DASHBOARD
==========================================
Total Records       : 10050
Total Machines      : 10
Normal Records      : 9721
Fault Records       : 329
Average Temperature : 47.70 °C
Average Vibration   : 2.47 mm/s
Average RPM         : 1403.22
==========================================
```

Screenshot 2: View Machine Records

```
CHOICE 02 – View Machine Records
Enter your choice: 2

======================================
          MACHINE RECORDS
======================================
9721 | IoT-M10 | 2026-09-08 12:00:00 | Temperature: 52.81 | Vibration: 1.874 | Fault: No Fault
7870 | IoT-M08 | 2026-09-10 01:15:00 | Temperature: 46.67 | Vibration: 2.401 | Fault: No Fault
7579 | IoT-M08 | 2026-09-07 00:30:00 | Temperature: 0.0 | Vibration: 2.421 | Fault: No Fault
7066 | IoT-M08 | 2026-09-01 16:15:00 | Temperature: 42.71 | Vibration: 2.904 | Fault: No Fault
4212 | IoT-M05 | 2026-09-03 04:45:00 | Temperature: 43.1 | Vibration: 2.332 | Fault: No Fault
5084 | IoT-M06 | 2026-09-01 20:45:00 | Temperature: 43.63 | Vibration: 3.768 | Fault: No Fault
1959 | IoT-M02 | 2026-09-10 23:30:00 | Temperature: 37.63 | Vibration: 2.683 | Fault: No Fault
1541 | IoT-M02 | 2026-09-06 15:00:00 | Temperature: 36.68 | Vibration: 2.398 | Fault: No Fault
4826 | IoT-M05 | 2026-09-09 14:15:00 | Temperature: 43.34 | Vibration: 2.659 | Fault: No Fault
5391 | IoT-M06 | 2026-09-05 01:30:00 | Temperature: 40.93 | Vibration: 3.318 | Fault: No Fault
3058 | IoT-M04 | 2026-09-01 14:15:00 | Temperature: 53.52 | Vibration: 1.829 | Fault: No Fault
409 | IoT-M01 | 2026-09-05 06:00:00 | Temperature: 50.76 | Vibration: 2.863 | Fault: No Fault
1126 | IoT-M02 | 2026-09-02 07:15:00 | Temperature: 37.82 | Vibration: 2.48 | Fault: No Fault
4149 | IoT-M05 | 2026-09-02 13:00:00 | Temperature: 42.23 | Vibration: 1.628 | Fault: No Fault
4001 | IoT-M05 | 2026-09-01 00:00:00 | Temperature: 41.24 | Vibration: 2.448 | Fault: No Fault
6929 | IoT-M07 | 2026-09-10 16:00:00 | Temperature: 50.19 | Vibration: 1.935 | Fault: No Fault
2964 | IoT-M03 | 2026-09-11 00:45:00 | Temperature: 47.3 | Vibration: 1.975 | Fault: No Fault
9244 | IoT-M10 | 2026-09-03 12:45:00 | Temperature: 51.41 | Vibration: 2.33 | Fault: No Fault
261 | IoT-M01 | 2026-09-03 17:00:00 | Temperature: 51.21 | Vibration: 2.671 | Fault: No Fault
4775 | IoT-M05 | 2026-09-09 01:30:00 | Temperature: 42.75 | Vibration: 2.739 | Fault: No Fault
--------------------------------------
Showing first 20 records.
```

CO-1: String Matching Algorithms

Screenshot 3: Search Machine Using KMP Algorithm

```
CHOICE 03 – Search Machine – KMP
Enter your choice: 3

Enter Machine ID:
Searching using KMP...
9721 | IoT-M10 | 2026-09-08 12:00:00 | Fault: No Fault
9244 | IoT-M10 | 2026-09-03 12:45:00 | Fault: No Fault
9111 | IoT-M10 | 2026-09-02 03:30:00 | Fault: No Fault
9069 | IoT-M10 | 2026-09-01 17:00:00 | Fault: No Fault
9042 | IoT-M10 | 2026-09-01 10:15:00 | Fault: No Fault
9152 | IoT-M10 | 2026-09-02 13:45:00 | Fault: No Fault
9448 | IoT-M10 | 2026-09-05 15:45:00 | Fault: No Fault
9580 | IoT-M10 | 2026-09-07 00:45:00 | Fault: No Fault
9980 | IoT-M10 | 2026-09-11 04:45:00 | Fault: No Fault
9455 | IoT-M10 | 2026-09-05 17:30:00 | Fault: No Fault
9485 | IoT-M10 | 2026-09-06 01:00:00 | Fault: No Fault
9420 | IoT-M10 | 2026-09-05 08:45:00 | Fault: No Fault
9511 | IoT-M10 | 2026-09-06 07:30:00 | Fault: No Fault
9158 | IoT-M10 | 2026-09-02 15:15:00 | Fault: No Fault
9063 | IoT-M10 | 2026-09-01 15:30:00 | Fault: No Fault
9751 | IoT-M10 | 2026-09-08 19:30:00 | Fault: No Fault
9507 | IoT-M10 | 2026-09-06 06:30:00 | Fault: No Fault
9626 | IoT-M10 | 2026-09-07 12:15:00 | Fault: No Fault
9698 | IoT-M10 | 2026-09-08 06:15:00 | Fault: No Fault
9519 | IoT-M10 | 2026-09-06 09:30:00 | Fault: No Fault

KMP search completed.
```

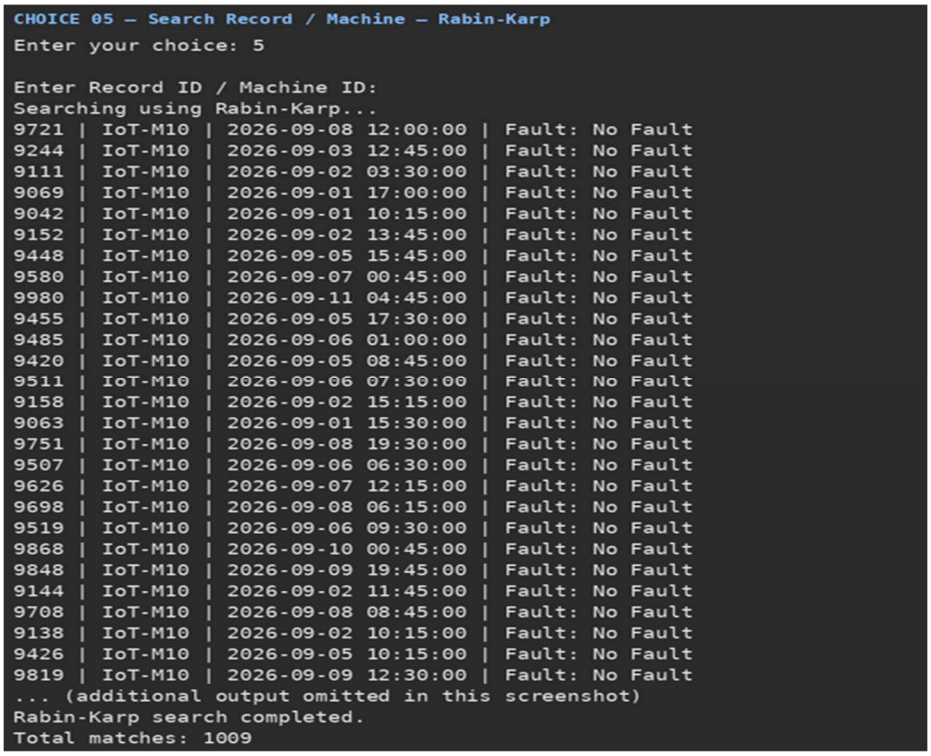
Screenshot 4: Search Fault Using Z-Function Algorithm

```
CHOICE 04 – Search Fault – Z-Function
Enter your choice: 4

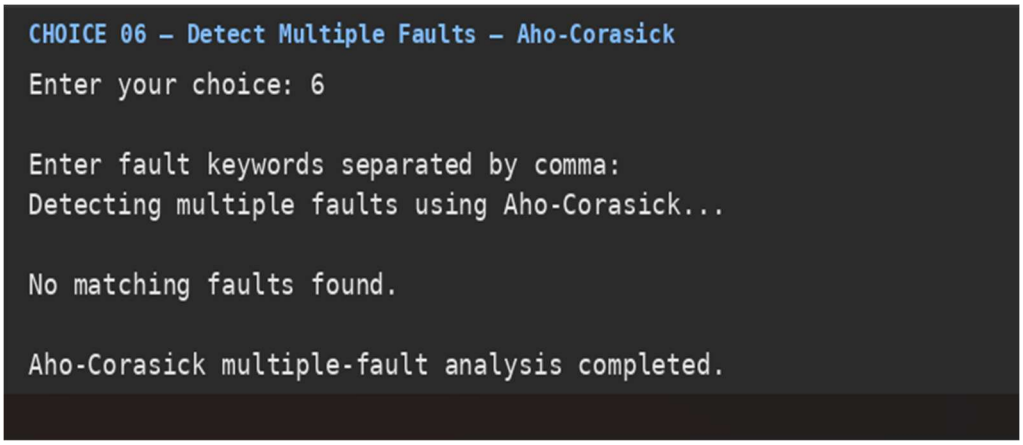
Enter fault type:
Searching using Z-Function...
9721 | IoT-M10 | 2026-09-08 12:00:00 | No Fault
7870 | IoT-M08 | 2026-09-10 01:15:00 | No Fault
7579 | IoT-M08 | 2026-09-07 00:30:00 | No Fault
7066 | IoT-M08 | 2026-09-01 16:15:00 | No Fault
4212 | IoT-M05 | 2026-09-03 04:45:00 | No Fault
5084 | IoT-M06 | 2026-09-01 20:45:00 | No Fault
1959 | IoT-M02 | 2026-09-10 23:30:00 | No Fault
1541 | IoT-M02 | 2026-09-06 15:00:00 | No Fault
4826 | IoT-M05 | 2026-09-09 14:15:00 | No Fault
5391 | IoT-M06 | 2026-09-05 01:30:00 | No Fault
3058 | IoT-M04 | 2026-09-01 14:15:00 | No Fault
409 | IoT-M01 | 2026-09-05 06:00:00 | No Fault
1126 | IoT-M02 | 2026-09-02 07:15:00 | No Fault
4149 | IoT-M05 | 2026-09-02 13:00:00 | No Fault
4001 | IoT-M05 | 2026-09-01 00:00:00 | No Fault
6929 | IoT-M07 | 2026-09-10 16:00:00 | No Fault
2964 | IoT-M03 | 2026-09-11 00:45:00 | No Fault
9244 | IoT-M10 | 2026-09-03 12:45:00 | No Fault
261 | IoT-M01 | 2026-09-03 17:00:00 | No Fault
4775 | IoT-M05 | 2026-09-09 01:30:00 | No Fault

Z-Function search completed.
```

Screenshot 5: Search Record / Machine Using Rabin-Karp Algorithm

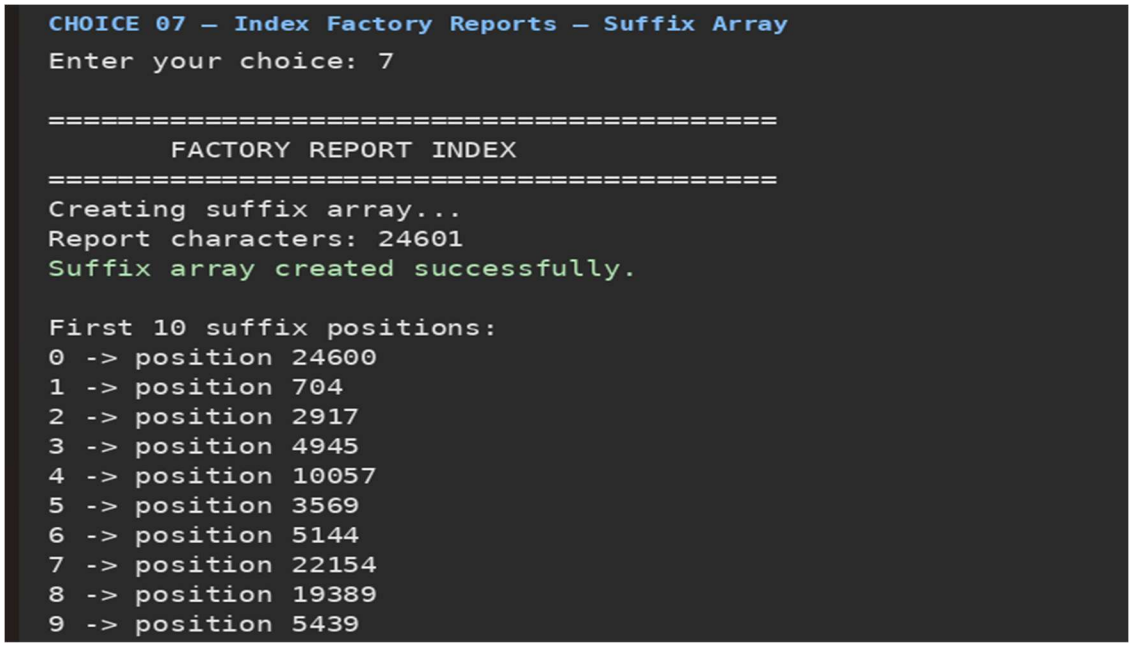


Screenshot 6: Detect Multiple Faults Using Aho-Corasick Algorithm

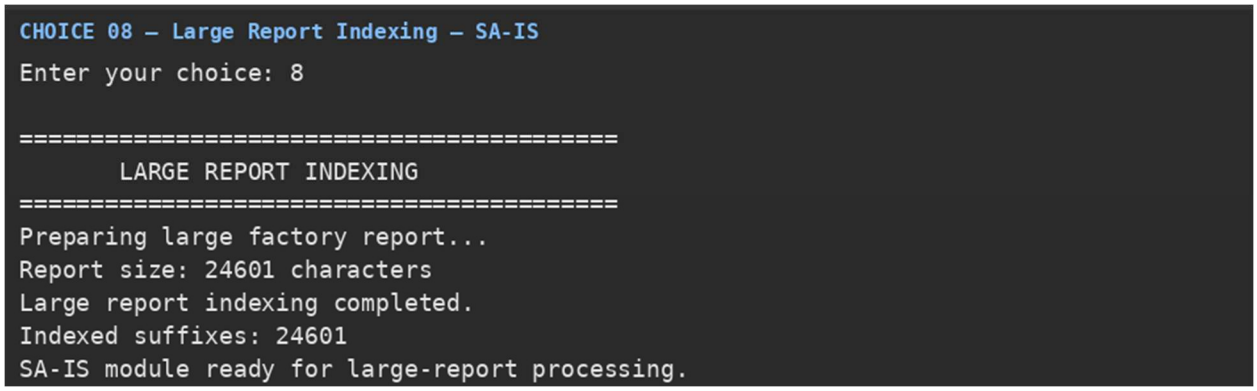


CO-2: Text Indexing Algorithms

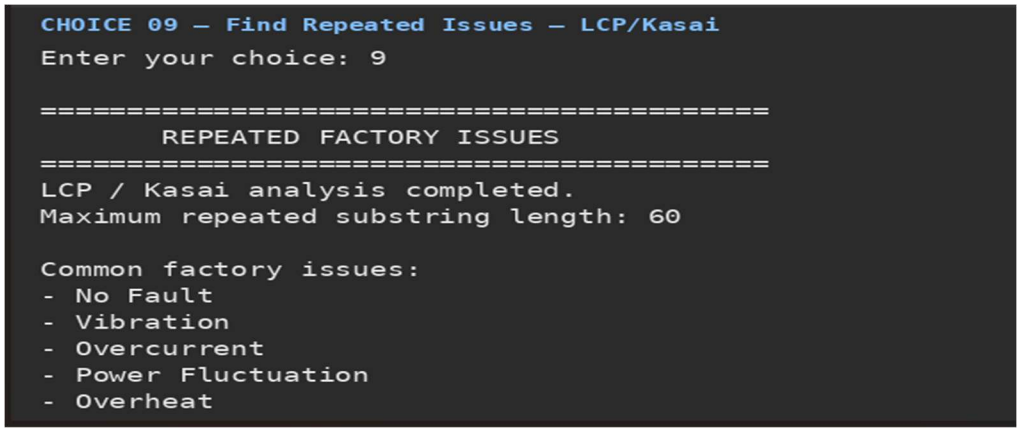
Screenshot 7: Index Factory Reports Using Suffix Array



Screenshot 8: Large Report Indexing Using SA-IS Algorithm



Screenshot 9: Find Repeated Issues Using LCP / Kasai Algorithm



Screenshot 10: Search Report Content Using Suffix Tree

```
CHOICE 10 – Search Report Content – Suffix Tree
Enter your choice: 10

Enter text to search in reports:
Searching using Suffix Tree...
Pattern not found.
```

Screenshot 11: Fast Report Search Using Suffix Automaton

```
CHOICE 11 – Fast Report Search – Suffix Automaton
Enter your choice: 11

Enter text to search:
Searching using Suffix Automaton...
Pattern not found.
Suffix Automaton search completed.
```

CO-3: Dynamic Programming Algorithms

Screenshot 12: Correct Machine Entry Using Levenshtein Distance

```
CHOICE 12 – Correct Machine Entry – Levenshtein
Enter your choice: 12

Enter Machine ID:
Suggested Machine ID: IoT-M10
Edit Distance: 0
```

Screenshot 13: Correct Typo Using Damerau-Levenshtein Distance

```
CHOICE 13 – Correct Typo – Damerau-Levenshtein
Enter your choice: 13

Enter Machine ID with possible typo:
Suggested correction: IoT-M10
Damerau-Levenshtein distance: 0
```

Screenshot 14: Validate Factory Data Using Weighted Edit Distance

```
CHOICE 14 – Validate Factory Data – Weighted Edit Distance
Enter your choice: 14

Enter Machine ID to validate:
Weighted validation completed.
Suggested value: IoT-M10
Weighted cost: 0
```

Screenshot 15: Optimize Processing Using Matrix-Chain Dynamic Programming

```
CHOICE 15 – Optimize Processing – Matrix-Chain DP
Enter your choice: 15

Enter number of processing stages: Enter 4 dimensions:

Minimum processing cost: 18000
Matrix-Chain DP optimization completed.
```

Screenshot 16: Select Machines Using Bitmask Dynamic Programming

```
CHOICE 16 – Select Machines – Bitmask DP
Enter your choice: 16

Enter required capacity/value:
Selected machines:
- IoT-M08 (Average RPM: 1403)
Machines selected: 1
Bitmask DP machine selection completed.
```

Screenshot 17: Plan Resources Using Subset Dynamic Programming

```
CHOICE 17 – Plan Resources – Subset DP
Enter your choice: 17

Enter number of resources required:
Selected resources:
- IoT-M09 (Cost: 127)
- IoT-M07 (Cost: 126)
- IoT-M10 (Cost: 147)
Minimum resource cost: 400
Subset DP resource planning completed.
```

CO-4: Network Flow Algorithms

Screenshot 18: Raw Material Flow Using Ford-Fulkerson Algorithm

```
CHOICE 18 — Raw Material Flow — Ford-Fulkerson
Enter your choice: 18

========================================
   RAW MATERIAL FLOW [FORD-FULKERSON]
========================================
Topology: Multi-Stage Industrial Plant (14 stations, 20 transfer links)
Source: Raw Material Hub (Source)
Sink  : Dispatch Warehouse (Sink)

Executing Ford-Fulkerson Augmenting Paths (DFS)...

Augmentation Trace (First 10 paths):
  Path #1: [Raw Material Hub (Source) -> Smelting & Casting Unit 1 -> CNC Center Alpha (M1) -> Sub-
    Assembly Line A -> Automated QA & Sensor Test -> Dispatch Warehouse (Sink)] | Added Flow: 36.48 units
  Path #2: [Raw Material Hub (Source) -> Smelting & Casting Unit 1 -> CNC Center Beta (M2) -> Sub-
    Assembly Line A -> Automated QA & Sensor Test -> Dispatch Warehouse (Sink)] | Added Flow: 16.84 units
  Path #3: [Raw Material Hub (Source) -> Smelting & Casting Unit 1 -> CNC Center Beta (M2) -> Sub-
    Assembly Line B -> Automated QA & Sensor Test -> Dispatch Warehouse (Sink)] | Added Flow: 14.03 units
  Path #4: [Raw Material Hub (Source) -> Forging & Stamping Unit 2 -> CNC Center Beta (M2) -> Sub-
    Assembly Line B -> Automated QA & Sensor Test -> Dispatch Warehouse (Sink)] | Added Flow: 5.61 units
  Path #5: [Raw Material Hub (Source) -> Forging & Stamping Unit 2 -> CNC Center Beta (M2) -> Sub-
    Assembly Line B -> Manual Quality Verification -> Dispatch Warehouse (Sink)] | Added Flow: 8.42 units
  Path #6: [Raw Material Hub (Source) -> Forging & Stamping Unit 2 -> Precision Lathe Gamma (M3) -> Sub-
    Assembly Line B -> Manual Quality Verification -> Dispatch Warehouse (Sink)] | Added Flow: 22.45
    units
  Path #7: [Raw Material Hub (Source) -> Material Prep & Ingot Milling -> Robotic Milling Delta (M4) ->
    Surface Treatment Cell -> Manual Quality Verification -> Dispatch Warehouse (Sink)] | Added Flow:
    30.87 units

----------------------------------------
Total Augmentation Iterations: 7
Maximum Material Flow Output : 134.71 units/hour
Ford-Fulkerson material flow modeling completed.
========================================
```

Screenshot 19: Production Link Maximum Flow Using Edmonds-Karp Algorithm

```
CHOICE 19 — Production Link Max Flow — Edmonds-Karp
Enter your choice: 19

========================================
  PRODUCTION LINK MAX FLOW [EDMONDS-KARP]
========================================
Executing Edmonds-Karp (BFS Shortest Augmenting Paths)...
Guarantees polynomial O(V * E^2) convergence.

Step-by-Step BFS Path Telemetry:
BFS Augmentation #1 (Length: 5 hops): Bottleneck = 36.48 units | Cumulative = 36.48
    Route: Raw Material Hub (Source) => Smelting & Casting Unit 1 => CNC Center Alpha (M1) => Sub-
    Assembly Line A => Automated QA & Sensor Test => Dispatch Warehouse (Sink)
BFS Augmentation #2 (Length: 5 hops): Bottleneck = 16.84 units | Cumulative = 53.32
    Route: Raw Material Hub (Source) => Smelting & Casting Unit 1 => CNC Center Beta (M2) => Sub-Assembly
     Line A => Automated QA & Sensor Test => Dispatch Warehouse (Sink)
BFS Augmentation #3 (Length: 5 hops): Bottleneck = 14.03 units | Cumulative = 67.35
    Route: Raw Material Hub (Source) => Smelting & Casting Unit 1 => CNC Center Beta (M2) => Sub-Assembly
     Line B => Automated QA & Sensor Test => Dispatch Warehouse (Sink)
BFS Augmentation #4 (Length: 5 hops): Bottleneck = 5.61 units | Cumulative = 72.97
    Route: Raw Material Hub (Source) => Forging & Stamping Unit 2 => CNC Center Beta (M2) => Sub-Assembly
     Line B => Automated QA & Sensor Test => Dispatch Warehouse (Sink)
BFS Augmentation #5 (Length: 5 hops): Bottleneck = 8.42 units | Cumulative = 81.39
    Route: Raw Material Hub (Source) => Forging & Stamping Unit 2 => CNC Center Beta (M2) => Sub-Assembly
     Line B => Manual Quality Verification => Dispatch Warehouse (Sink)
BFS Augmentation #6 (Length: 5 hops): Bottleneck = 22.45 units | Cumulative = 103.84
    Route: Raw Material Hub (Source) => Forging & Stamping Unit 2 => Precision Lathe Gamma (M3) => Sub-
    Assembly Line B => Manual Quality Verification => Dispatch Warehouse (Sink)
BFS Augmentation #7 (Length: 5 hops): Bottleneck = 30.87 units | Cumulative = 134.71
    Route: Raw Material Hub (Source) => Material Prep & Ingot Milling => Robotic Milling Delta (M4) =>
    Surface Treatment Cell => Manual Quality Verification => Dispatch Warehouse (Sink)

----------------------------------------
Total Shortest Augmentations : 7
Maximum Production Flow Rate : 134.71 units/hour
Edmonds-Karp production link optimization completed.
========================================
```

Screenshot 20: Large Factory Network Flow Using Dinic's Algorithm

```
CHOICE 20 — Large Factory Network Flow — Dinic
Enter your choice: 20

Enter number of manufacturing stages (e.g. 4): Enter number of parallel units per stage (e.g. 4):
========================================
   LARGE FACTORY FLOW [DINIC'S ALGORITHM]
========================================
Generated Network: 4 Stages x 4 Parallel Units (18 Total Stations, 32 Directed Links)
Executing Dinic's Algorithm (Level Graph BFS + Blocking Flow DFS)...

Execution Results:
  Level-Graph BFS Phases : 2 phases
  Execution Time (Nano)  : 740,610 ns (0.741 ms)
  Total Maximum Flow     : 252.00 units/hour
Dinic's algorithm completed successfully for large network.
========================================
```

Screenshot 21: Worker-Machine Assignment Using Bipartite Matching

```
CHOICE 21 — Worker-Machine Assignment — Bipartite Matching
Enter your choice: 21

========================================
 WORKER-MACHINE ASSIGNMENT [BIPARTITE MATCHING]
========================================
Active Factory Workforce   : 7 technicians
Available Production Units : 7 machines

Computing Maximum Bipartite Matching...

Optimal Worker -> Machine Assignments (7 Total Matches):
  [ASSIGNED] Elena Rostova (Robotics Tech)      ===> Robotic Milling Delta (M4)
  [ASSIGNED] Farhan Malik (Stamping Master)     ===> Stamping Unit (M5)
  [ASSIGNED] Alex Rivera (CNC Specialist)       ===> CNC Alpha (M1)
  [ASSIGNED] Dana White (QA Analyst)            ===> Automated QA Cell (M6)
  [ASSIGNED] Grace Hopper (System Overseer)     ===> Thermal Treatment Cell (M7)
  [ASSIGNED] Blake Chen (Milling Tech)          ===> CNC Beta (M2)
  [ASSIGNED] Carlos Gomez (Lathe Operator)      ===> Precision Lathe Gamma (M3)
Bipartite matching worker allocation completed.
========================================
```

Screenshot 22: Bottleneck Identification Using Max-Flow Min-Cut Theorem

```
CHOICE 22 — Bottleneck Identification — Max-Flow Min-Cut
Enter your choice: 22

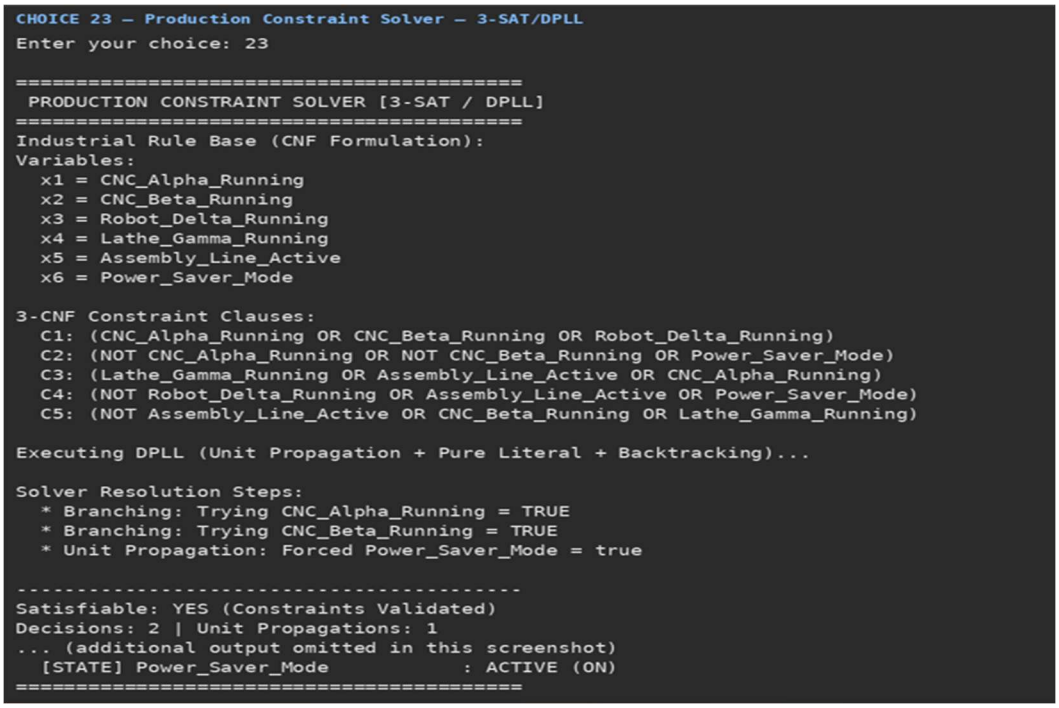
========================================
  BOTTLENECK IDENTIFICATION [MAX-FLOW MIN-CUT]
========================================
Factory Maximum Flow Throughput: 134.71 units/hour
Minimum Cut Capacity           : 134.71 units/hour
Max-Flow == Min-Cut Duality Verified: YES (Optimal Cut)

CRITICAL PRODUCTION BOTTLENECKS (Saturated Cut Edges):
  1. Automated QA & Sensor Test [Stage] -> Dispatch Warehouse (Sink) [Sink]
     Link: Auto QA to Dispatch | Flow: 73.0 / Cap: 73.0 (100.0% Saturated)
  2. Manual Quality Verification [Stage] -> Dispatch Warehouse (Sink) [Sink]
     Link: Manual QA to Dispatch | Flow: 61.7 / Cap: 61.7 (100.0% Saturated)

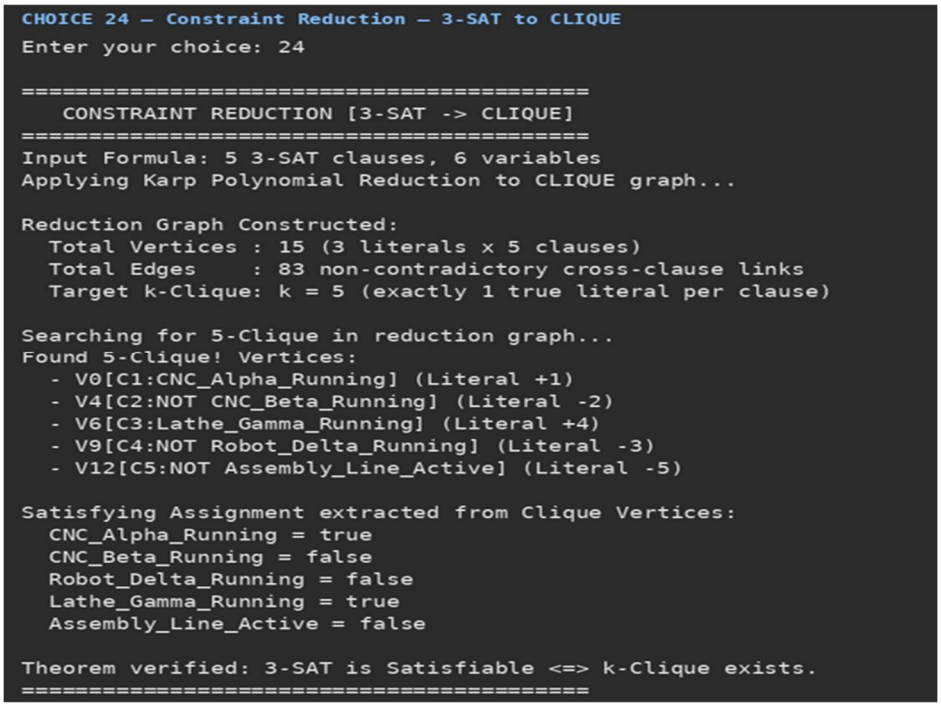
ENGINEERING RECOMMENDATION:
Expanding capacity on the above cut links will directly increase
overall plant throughput. Expanding non-cut links will have ZERO impact!
========================================
```

CO-5: NP-Completeness and Approximation Algorithms

Screenshot 23: Production Constraint Solver Using 3-SAT / DPLL



Screenshot 24: Constraint Reduction from 3-SAT to CLIQUE



Screenshot 25: Conflict Activity Reduction Using CLIQUE, Independent Set and Vertex Cover

```
CHOICE 25 — Conflict Activity Reduction
Enter your choice: 25

========================================
 CONFLICT REDUCTION [CLIQUE -> IS -> VC]
========================================
Manufacturing Operations & Shared Resource Conflicts:
  [0] Milling-01 [High Power, Tool 1]
  [1] Turning-02 [Tool 1, Gantry Crane]
  [2] Welding-03 [Gantry Crane, Shield Gas]
  [3] Hardening-04 [High Power, Cooling Loop]
  [4] Inspection-05 [Shield Gas, Clean Zone]
  [5] Stamping-06 [High Power, Tool 2]

Conflict Links (Resource Contention): 6 active edges
  * Conflict: Milling-01 [High Power, Tool 1] <===> Turning-02 [Tool 1, Gantry Crane]
  * Conflict: Milling-01 [High Power, Tool 1] <===> Hardening-04 [High Power, Cooling Loop]
  * Conflict: Milling-01 [High Power, Tool 1] <===> Stamping-06 [High Power, Tool 2]
  * Conflict: Turning-02 [Tool 1, Gantry Crane] <===> Welding-03 [Gantry Crane, Shield Gas]
  * Conflict: Welding-03 [Gantry Crane, Shield Gas] <===> Inspection-05 [Shield Gas, Clean Zone]
  * Conflict: Hardening-04 [High Power, Cooling Loop] <===> Stamping-06 [High Power, Tool 2]

Step 1: CLIQUE on Compatibility Graph <==> INDEPENDENT SET on Conflict Graph
An Independent Set represents mutually non-conflicting tasks that CAN execute simultaneously.

Maximum Independent Set (Size = 3 compatible operations):
  [SCHEDULED] Turning-02 [Tool 1, Gantry Crane]
  [SCHEDULED] Hardening-04 [High Power, Cooling Loop]
  [SCHEDULED] Inspection-05 [Shield Gas, Clean Zone]

Step 2: INDEPENDENT SET <==> VERTEX COVER Reduction
Theorem: S is an Independent Set <=> V \ S is a Vertex Cover.
... (additional output omitted in this screenshot)
Reduction duality validated: Max-IS size + Min-VC size = |V| (6 == 6).
========================================
```

Screenshot 26: Conflict Resolution Using Vertex Cover 2-Approximation

```
CHOICE 26 — Conflict Resolution — Vertex Cover 2-Approximation
Enter your choice: 26

========================================
 CONFLICT RESOLUTION [VERTEX-COVER 2-APPROX]
========================================
Factory Conflict Graph: 6 tasks, 6 conflict edges.
Computing 2-Approximation using Maximal Matching Edge Selection...

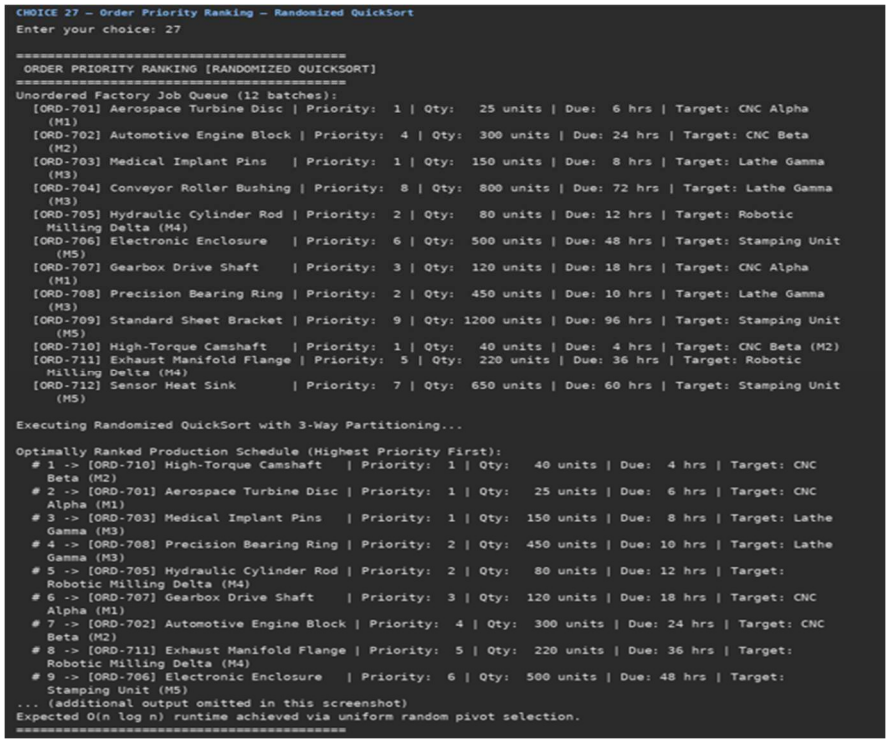
Maximal Disjoint Edge Matching Chosen (M):
  Edge: [Milling-01 [High Power, Tool 1]] <---> [Turning-02 [Tool 1, Gantry Crane]]
  Edge: [Welding-03 [Gantry Crane, Shield Gas]] <---> [Inspection-05 [Shield Gas, Clean Zone]]
  Edge: [Hardening-04 [High Power, Cooling Loop]] <---> [Stamping-06 [High Power, Tool 2]]

2-Approximation Vertex Cover Result (Size: 6):
  [COVER NODE] Task 0: Milling-01 [High Power, Tool 1]
  [COVER NODE] Task 1: Turning-02 [Tool 1, Gantry Crane]
  [COVER NODE] Task 2: Welding-03 [Gantry Crane, Shield Gas]
  [COVER NODE] Task 3: Hardening-04 [High Power, Cooling Loop]
  [COVER NODE] Task 4: Inspection-05 [Shield Gas, Clean Zone]
  [COVER NODE] Task 5: Stamping-06 [High Power, Tool 2]

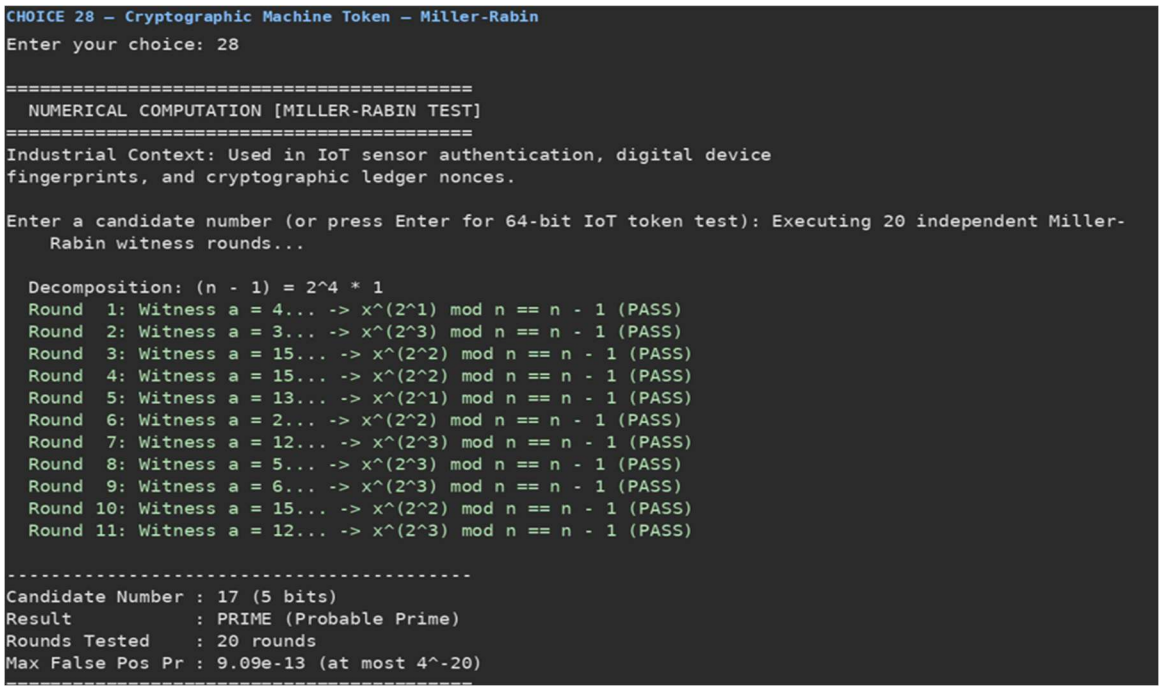
Approximation Guarantee Analysis:
  Maximal Matching Size |M| : 3
  Approx Cover Size |C|     : 6  (= 2 * |M|)
  Since any valid cover must touch at least 1 endpoint of each disjoint matched edge,
  Optimal |C*| >= 3. Hence: |C| <= 2 * |C*| (Guaranteed within 2x optimal).
Vertex Cover 2-approximation completed successfully.
========================================
```

CO-6: Randomized and Parallel Algorithms

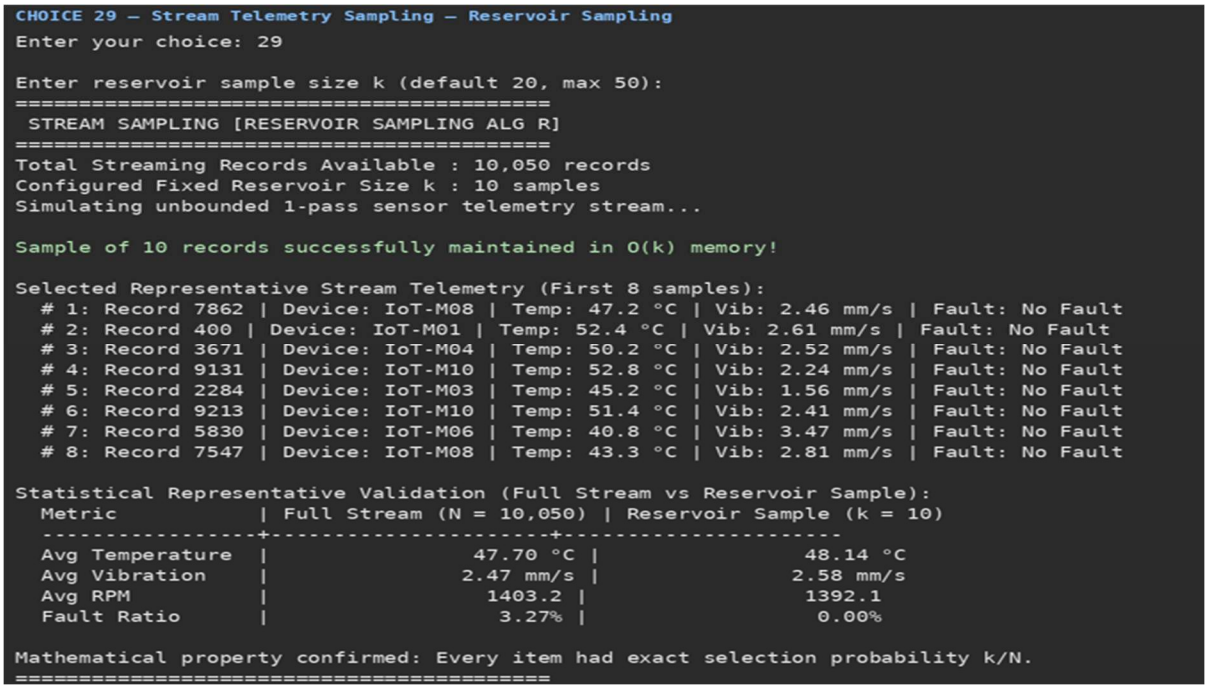
Screenshot 27: Order Priority Ranking Using Randomized QuickSort



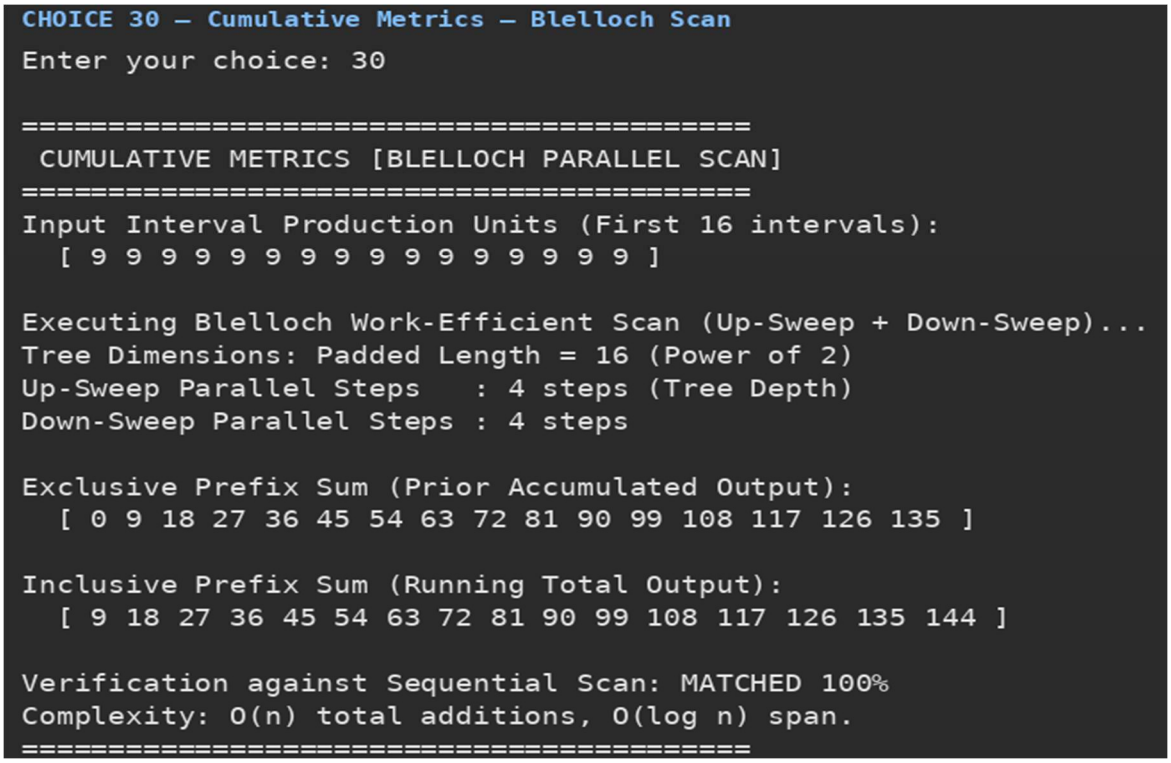
Screenshot 28: Cryptographic Machine Token Using Miller-Rabin Primality Test



Screenshot 29: Stream Telemetry Sampling Using Reservoir Sampling



Screenshot 30: Cumulative Metrics Using Blelloch Work-Efficient Scan



Screenshot 31: Large-Scale Data Aggregation Using Parallel Tree Reduction

```
CHOICE 31 — Large-Scale Data Aggregate — Parallel Tree Reduce
Enter your choice: 31

========================================
 DATA AGGREGATION [PARALLEL TREE REDUCE]
========================================
Aggregating Entire Factory Dataset (10,050 sensor telemetry records)...
Executing ForkJoin Parallel Reduction across available CPU cores...

Consolidated Plant Metrics:
  Total Records Processed : 10,050
  Total Electrical Power  : 24,065.24 kW
  Maximum Temperature     : 999.00 °C
  Peak Vibration Recorded : 8.92 mm/s
  Mean Operating RPM      : 1403.22 RPM
  Total Machine Faults    : 329 records

Multi-Core Parallel Performance Benchmark:
  Available Hardware Cores: 2 CPU cores
  Sequential Single-Thread: 51,199,040 ns (51.199 ms)
  Parallel ForkJoin Tree  : 10,412,958 ns (10.413 ms)
  Measured Speedup Factor : 4.92x
========================================
```

Screenshot 32: Scalability Evaluation Using Brent's Theorem

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CHOICE 32 — Scalability Evaluation — Brent's Theorem
Enter your choice: 32

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 PARALLEL EVALUATION [BRENT'S THEOREM]
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Brent's Work-Time Scheduling Principle: T_p <= (T_1 - T_inf) / p + T_inf

Workload Profile: Real-Time Sensor Tree Reduction
  Model Formula : Tree Reduction: T1 = n - 1, T_inf = ceil(log2(n))
  Total Work T1 : 10,049 operations (1 processor)
  Span T_inf    : 14 critical path depth (inf processors)
  Half-Perf p1/2: 717 processors (T1 / T_inf)

  Scaling Table across Processor Counts (p):
  ------------------------------------------------------------------------
  Cores (p) | Brent Time T_p | Lower Bound | Speedup S_p | Efficiency E_p
  ----------+----------------+-------------+-------------+----------------
          1 |        10049.0 |     10049.0 |       1.00x |          100.0%
          2 |         5031.5 |      5024.5 |       2.00x |           99.9%
          4 |         2522.8 |      2512.3 |       3.98x |           99.6%
          8 |         1268.4 |      1256.1 |       7.92x |           99.0%
         16 |          641.2 |       628.1 |      15.67x |           98.0%
         32 |          327.6 |       314.0 |      30.68x |           95.9%
         64 |          170.8 |       157.0 |      58.84x |           91.9%
        128 |           92.4 |        78.5 |     108.76x |           85.0%
  ------------------------------------------------------------------------

Workload Profile: Cumulative Production Blelloch Scan
  Model Formula : Work-Efficient Scan: T1 = 2n, T_inf = 2*log2(n)
  Total Work T1 : 20,100 operations (1 processor)
  Span T_inf    : 28 critical path depth (inf processors)
... (additional output omitted in this screenshot)
Brent's Theorem evaluation completed.
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